

Documentation for the 3D Model Pack

Acknowledgements

Hello, and thank you for purchasing my 3D model pack. I would like to mention that this is the first time I have created and sold such a pack. Please feel free to leave a review on the Unity Asset Store if you enjoyed this pack. Your feedback would help support and encourage me to improve further and create even higher quality packs.

Introduction

To help you understand how the various 3D models work, I recommend opening the demonstration scene included in this pack.

Pack contents

This pack includes **28 3D models, all of which are animated, scripted and with sound** :

- 2 Buttons On Stand
- 2 Simple Switches
- 3 Levers
- 3 Toggle Switches
- 1 Toggle
- 1 Bidirectional Panel (up and down arrow)
- 4 Geometric Buttons
- 2 Valves
- 5 Square Pressure Plates
 - o Including 1 without a symbol and 4 with different symbols: heart, electricity, star, skull
- 5 Round Pressure Plates
 - o Including 1 without a symbol and 4 with different symbols: heart, electricity, star, skull

There are **8 distinct sounds that harmonize well with the 28 3D models and 8 scripts distributed across the different 3D models.**

Techniques used for this pack

The method used to detect the player is "OnTriggerEnter()".

The prefabs are designed so that you only need to complete the action to be performed inside the script, whether it's by interacting with a button, a lever, etc.

Pressure plates and valves are animated by script.

You can, of course, use your own scripts, animations, and sounds if you wish.

How to use the prefabs included in this pack

To use a prefab, you must first assign the "Player" tag to your character and add a Rigidbody to it.

You can also use the 'CapsulePlayer' prefab, which is a capsule character configured to interact with the other 3D models in the pack.

Then, drag the prefab and press "E" when your character is in the trigger zone (represented by the parent element's Collider), and you're done!

Specificity of the pack

The Bidirectional Panel works a bit differently from the other objects because the buttons are quite close together. Therefore, I use the concept of "Raycast" to send a ray from the player and detect which button is being looked at.

To use the Bidirectional Panel, drag its prefab and link your character's camera to its associated script.

The Valves are also quite different. For the valves, you need to apply one of the two scripts:

- LightValveController
- ParticleValveController

The "LightValveController" script allows you to control a valve object that affects the intensity of a light source. Turning the valve to the left will decrease the light intensity, while turning it to the right will increase it. You can customize everything about the valves: maximum rotation angle (360 = 1 full rotation), minimum and maximum intensity, current intensity (at project start), and the valve's rotation speed. The script will adjust the valve's rotation angle proportionally to the current intensity.

The principle is the same with the "ParticleValveController" script, except that it works with particles.

To use the Handle Valve or the Circular Valve, drag its prefab and your particle system onto the ParticleValveController script associated with the prefab. If you want to use the LightValveController script, simply remove the other script from your prefab and replace it.

If you want the valves to control something else, you'll have very little to change. Additionally, the lines to modify are specified in each of the two valve management scripts.

Additional information

Updates and improvements to this pack are possible in the future.